In dynamic microphone, between the two-strong magnetic ends a coil spring is placed. A diaphragm is attached along with this coil spring. The diaphragm is firmly attached to the body of the microphone. The coil spring laminated so it won't have any contact with other parts. This total setup is fixed in a case.

When sound waves strike the diaphragm, the coil attached to it will vibrate front and back. Due to this, the coil cuts the magnetic field and creating electrical signal. The strength of this electrical signal and frequency decides the true deflection of vibration occurred in the diaphragm. This electrical signal is the output of the microphone. The dynamic microphone will give equal frequency response for the sound waves in the range of 50 Hz to 10000 Hz. It is made up of low impedance material.

2.4.1.2 Condenser Microphone

When the distance between the electrodes of a capacitor gets changed the capacitance also varies. The condenser microphone is working under this principle. A condenser microphone is shown in Figure 2.25. The construction of this microphone is also similar to the construction of a capacitor. In this, one plate is kept static and another plate can vibrate on receiving the sound wave.

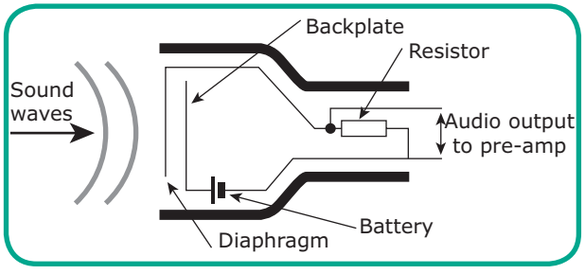


Figure 2.25 Condenser Microphone

The vibrating plate will act as diaphragm. When the sound signal strikes the microphone, the vibrating plate starts slight movement (vibration) in front and back. Due to this, the gap between the two plates gets changed and in turn there is a change in capacitance occurs. The change in capacitance depends upon the sound waves. Hence, the electrical signal produced also gets changed and is taken as output. The condenser microphone is a high impedance microphone.

2.4.2 Loud Speaker

A material which converts electrical energy into mechanical energy and converting such energy into sound energy is known as loud speaker. Simply to say, the part that converts amplified audio frequency signals back to the original sound is the speaker.

2.4.2.1 Dynamic Loudspeaker

Figure 2.26 shows a cross sectional view of a Dynamic Loudspeaker.

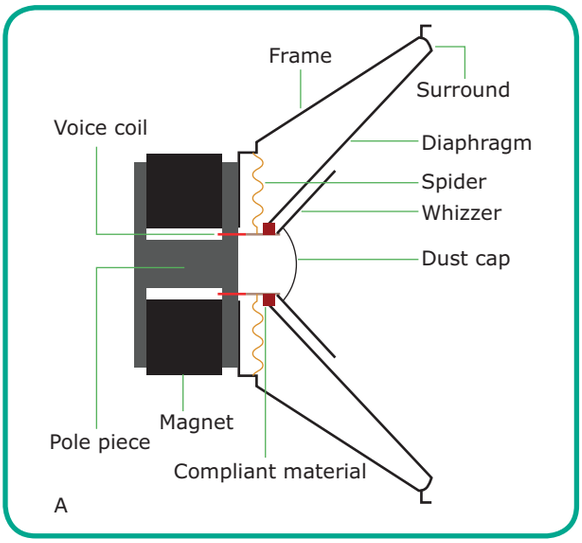


Figure 2.26 Dynamic Loudspeaker

Voice coil is placed in between the strong magnetic ends in order to fill the air gap. Spider is placed on the voice coil.

Spider helps the voice coil to fill the air gap. Paper cone is used to connect the

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